# Chapter 1. Preliminaries to Complex Analysis

Complex numbers, holomorphic functions, power series, integration along curves

## Contents

**1 Complex numbers and the complex plane** **2**

1.1 Basic properties . . . . . 2

1.2 Convergence . . . . . 2

1.3 Sets in the complex plane . . . . . 2

**2 Functions on the complex plane** **3**

2.1 Continuous functions . . . . . 3

2.2 Holomorphic functions . . . . . 4

2.3 Power series . . . . . 6

**3 Integration along curves** **7**

# Overview

Complex numbers, the topology of $\mathbb{C}$, holomorphic functions (Cauchy–Riemann equations, power series), and curve integrals: if $f$ has a primitive, $\int_\gamma f(z)\,dz = 0$ for every closed $\gamma$.

# 1 Complex numbers and the complex plane

## 1.1 Basic properties

A complex number is $z = x + iy$, $x, y \in \mathbb{R}$, $i^2 = -1$, with **real part** $x = \mathrm{Re}(z)$ and **imaginary part** $y = \mathrm{Im}(z)$; $z$ is **purely imaginary** if $\mathrm{Re}\, z = 0$. $\mathbb{C}$ is identified with $\mathbb{R}^2$ via $z \leftrightarrow (x, y)$, with **real axis** and **imaginary axis**.

Treating all numbers as real and using $i^2 = -1$: if $z_1 = x_1 + iy_1$, $z_2 = x_2 + iy_2$,

$$z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2), \qquad z_1 z_2 = (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + y_1 x_2).$$

These operations are commutative, associative and distributive. Addition is vector addition; multiplication by $i$ is rotation by $\pi/2$.

The **absolute value** of $z = x + iy$ is $|z| = (x^2 + y^2)^{1/2}$, the euclidean distance from 0 to $(x, y)$. The triangle inequality $|z + w| \le |z| + |w|$ holds, and also $|\mathrm{Re}(z)| \le |z|$, $|\mathrm{Im}(z)| \le |z|$ and

$$\big||z| - |w|\big| \le |z - w|,$$

which follows from $|z| \le |z - w| + |w|$ and $|w| \le |z - w| + |z|$.

The **complex conjugate** of $z = x + iy$ is $\overline{z} = x - iy$ (reflection across the real axis). $z$ is real iff $z = \overline{z}$, purely imaginary iff $z = -\overline{z}$, and

$$\mathrm{Re}(z) = \frac{z + \overline{z}}{2}, \qquad \mathrm{Im}(z) = \frac{z - \overline{z}}{2i}, \qquad |z|^2 = z\overline{z}, \qquad \frac{1}{z} = \frac{\overline{z}}{|z|^2}\ (z \neq 0).$$

Any $z \neq 0$ has a **polar form** $z = re^{i\theta}$, $r = |z| > 0$, $e^{i\theta} = \cos\theta + i\sin\theta$; $\theta$ is the **argument** $\arg z$ (defined mod $2\pi$), the counterclockwise angle from the positive real axis. If $z = re^{i\theta}$, $w = se^{i\varphi}$, then $zw = rse^{i(\theta+\varphi)}$: multiplication is a rotation composed with a dilation.

## 1.2 Convergence

$\{z_n\}$ **converges** to $w$, $w = \lim z_n$, if $|z_n - w| \to 0$, i.e. the points converge in $\mathbb{R}^2$, equivalently the real and imaginary parts converge. $\{z_n\}$ is a **Cauchy sequence** if $|z_n - z_m| \to 0$ as $n, m \to \infty$. $\{z_n\}$ is Cauchy iff its real and imaginary parts are, and $\mathbb{R}$ is complete; hence:

**Theorem 1.1**

$\mathbb{C}$, the complex numbers, is complete.

## 1.3 Sets in the complex plane

For $z_0 \in \mathbb{C}$, $r > 0$: the **open disc** $D_r(z_0) = \{|z - z_0| < r\}$, the **closed disc** $\overline{D}_r(z_0) = \{|z - z_0| \le r\}$, the circle $C_r(z_0) = \{|z - z_0| = r\}$, and the unit disc $\mathbb{D} = \{|z| < 1\}$.

For $\Omega \subset \mathbb{C}$: $z_0$ is an **interior point** if $D_r(z_0) \subset \Omega$ for some $r > 0$, and the **interior** is the set of these. $\Omega$ is **open** if all its points are interior, **closed** if $\Omega^c = \mathbb{C} - \Omega$ is open. $z$ is a **limit point** if $z_n \to z$ for some $z_n \in \Omega$, $z_n \neq z$; closed sets are those containing their limit points. The **closure** $\overline{\Omega}$

adds the limit points; the **boundary** $\partial\Omega$ is closure minus interior. $\Omega$ is **bounded** if $|z| < M$ on $\Omega$, with $\text{diam}(\Omega) = \sup_{z,w\in\Omega}|z-w|$, and **compact** if closed and bounded.

**Theorem 1.2**

The set $\Omega \subset \mathbb{C}$ is compact if and only if every sequence $\{z_n\} \subset \Omega$ has a subsequence that converges to a point in $\Omega$.

**Proof [Supplement].** If $\Omega$ is closed and bounded, Bolzano–Weierstrass (twice) gives a convergent subsequence, with limit in $\Omega$ since $\Omega$ is closed. Conversely, if $\Omega$ were unbounded, $z_n \in \Omega$ with $|z_n| > n$ has no convergent subsequence; if not closed, a limit point $z \notin \Omega$ is the limit of $z_n \in \Omega$, and every subsequence converges to $z \notin \Omega$. ■

An **open covering** of $\Omega$ is a family of open sets $\{U_\alpha\}$ with $\Omega \subset \bigcup_\alpha U_\alpha$.

**Theorem 1.3**

A set $\Omega$ is compact if and only if every open covering of $\Omega$ has a finite subcovering.

**Proof [Supplement].** Suppose $\Omega$ is compact and the open covering $\{U_\alpha\}$ has no finite subcovering. Put $\Omega$ in a square $Q_0$ and quarter it; for some quarter $Q_1$, $\Omega \cap Q_1$ is not finitely covered. Repeating gives nested $Q_n$, $\text{diam}(Q_n) \to 0$, with $\Omega \cap Q_n$ not finitely covered. Points $z_n \in \Omega \cap Q_n$ form a Cauchy sequence with limit $w \in \Omega$, $w \in U_\alpha$ for some $\alpha$; as $U_\alpha$ is open, $Q_n \subset U_\alpha$ for large $n$, a contradiction. Conversely, if every open covering has a finite subcovering, the discs $D_n(0)$ show $\Omega$ is bounded, and a limit point $z \notin \Omega$ would make $\{w : |w-z| > 1/n\}$ a covering with no finite subcovering. ■

**Proposition 1.4**

If $\Omega_1 \supset \Omega_2 \supset \cdots$ are non-empty compact sets with $\text{diam}(\Omega_n) \to 0$, there is a unique $w \in \mathbb{C}$ with $w \in \Omega_n$ for all $n$.

**Proof.** Choose $z_n \in \Omega_n$; $\text{diam}(\Omega_n) \to 0$ makes $\{z_n\}$ Cauchy, with limit $w$. Each $\Omega_n$ is closed and contains $z_m$ for $m \geq n$, so $w \in \Omega_n$. A second such $w' \neq w$ would contradict $\text{diam}(\Omega_n) \to 0$. ■

An open $\Omega$ is **connected** if it is not a union of two disjoint non-empty open sets; a connected open set is a **region**. A closed $F$ is connected if it is not a union of two disjoint non-empty closed sets. An open $\Omega$ is connected iff any two of its points can be joined by a curve in $\Omega$.

# 2 Functions on the complex plane

## 2.1 Continuous functions

$f$ on $\Omega$ is **continuous** at $z_0$ if for every $\epsilon > 0$ there is $\delta > 0$ with $|f(z) - f(z_0)| < \epsilon$ for $z \in \Omega$, $|z - z_0| < \delta$; equivalently $z_n \to z_0$ implies $f(z_n) \to f(z_0)$. Sums and products of continuous functions are continuous; $f$ is continuous iff it is continuous in $(x, y)$; then $|f|$ is continuous. $f$ **attains a maximum** at $z_0$ if $|f(z)| \leq |f(z_0)|$ on $\Omega$ (reversed for a minimum).

**Theorem 2.1**

A continuous function on a compact set $\Omega$ is bounded and attains a maximum and minimum on $\Omega$.

(The proof is the same as for functions of a real variable.)

## 2.2 Holomorphic functions

**Definition (holomorphic function)**

Let $\Omega$ be open and $f$ complex-valued on $\Omega$. $f$ is **holomorphic at** $z_0 \in \Omega$ if the quotient

$$\frac{f(z_0+h)-f(z_0)}{h} \tag{1}$$

converges as $h \to 0$ ($h \in \mathbb{C}-\{0\}$, $z_0+h \in \Omega$); the limit is the **derivative** $f'(z_0)$. $f$ is **holomorphic on** $\Omega$ if holomorphic at every point; on a closed $C$, if holomorphic on an open set containing $C$; **entire** if holomorphic on $\mathbb{C}$.

Here $h$ approaches 0 from any direction. (Also: *regular*, *complex differentiable*.) Holomorphic functions turn out to be infinitely differentiable and *analytic*, unlike real-differentiable ones.

**Example 1**

$f(z)=z$ is holomorphic on any open set with $f'(z)=1$; any polynomial $p(z)=a_0+a_1z+\cdots+a_nz^n$ is entire, with $p'(z)=a_1+\cdots+na_nz^{n-1}$ (by Proposition 2.2).

**Example 2**

$1/z$ is holomorphic on any open set not containing 0, with $f'(z)=-1/z^2$.

**Example 3**

$f(z)=\overline{z}$ is not holomorphic: $\dfrac{f(z_0+h)-f(z_0)}{h}=\dfrac{\overline{h}}{h}$, which has no limit as $h\to 0$ (it is 1 for real $h$ and $-1$ for purely imaginary $h$).

Clearly $f$ is holomorphic at $z_0$ iff there is $a\in\mathbb{C}$ with

$$f(z_0+h)-f(z_0)-ah=h\psi(h), \tag{2}$$

where $\psi(h)\to 0$ as $h\to 0$; then $a=f'(z_0)$. So holomorphic functions are continuous. As for one real variable, using (2):

**Proposition 2.2**

If $f$ and $g$ are holomorphic in $\Omega$, then:
(i) $f+g$ is holomorphic in $\Omega$ and $(f+g)'=f'+g'$.
(ii) $fg$ is holomorphic in $\Omega$ and $(fg)'=f'g+fg'$.
(iii) If $g(z_0)\neq 0$, then $f/g$ is holomorphic at $z_0$ and $(f/g)'=\dfrac{f'g-fg'}{g^2}$. Moreover, if $f:\Omega\to U$ and $g:U\to\mathbb{C}$ are holomorphic, the chain rule holds: $(g\circ f)'(z)=g'(f(z))f'(z)$ for all $z\in\Omega$.

**Proof [Supplement](chain rule).** Write $f(z+h)-f(z)=f'(z)h+h\psi_1(h)$ and $g(w+k)-g(w)=g'(w)k+k\psi_2(k)$ with $\psi_j\to 0$, $w=f(z)$. With $k=f(z+h)-f(z)\to 0$, $g(f(z+h))-g(f(z))=g'(w)k+k\psi_2(k)=g'(w)f'(z)h+h[g'(w)\psi_1(h)+(f'(z)+\psi_1(h))\psi_2(k)]$, and the bracket tends to 0. (i)–(iii) are similar. ■

**Complex-valued functions as mappings.** $\overline{z}$ corresponds to the linear map $F:(x,y)\mapsto(x,-y)$, real-differentiable to all orders: real differentiability does not imply holomorphy. To $f=u+iv$ associate $F(x,y)=(u(x,y),v(x,y))$. $F$ is **differentiable** at $P_0=(x_0,y_0)$ if there is a linear $J:\mathbb{R}^2\to\mathbb{R}^2$ with

$$\frac{|F(P_0+H)-F(P_0)-J(H)|}{|H|}\to 0 \quad \text{as } |H|\to 0,\ H\in\mathbb{R}^2; \tag{3}$$

then $J$ is the Jacobian $J_F=\begin{pmatrix}\partial u/\partial x & \partial u/\partial y\\ \partial v/\partial x & \partial v/\partial y\end{pmatrix}$. If $f$ is holomorphic at $z_0$, real $h=h_1$ in (1) gives $f'(z_0)=\frac{\partial f}{\partial x}(z_0)$ and $h=ih_2$ gives $f'(z_0)=\frac{1}{i}\frac{\partial f}{\partial y}(z_0)$. So $\frac{\partial f}{\partial x}=\frac{1}{i}\frac{\partial f}{\partial y}$; real and imaginary parts give the **Cauchy–Riemann equations**

$$\frac{\partial u}{\partial x}=\frac{\partial v}{\partial y}\qquad\text{and}\qquad\frac{\partial u}{\partial y}=-\frac{\partial v}{\partial x}.$$

Define the operators

$$\frac{\partial}{\partial z}=\frac{1}{2}\Big(\frac{\partial}{\partial x}+\frac{1}{i}\frac{\partial}{\partial y}\Big),\qquad \frac{\partial}{\partial \overline{z}}=\frac{1}{2}\Big(\frac{\partial}{\partial x}-\frac{1}{i}\frac{\partial}{\partial y}\Big).$$

**Proposition 2.3**

If $f$ is holomorphic at $z_0$, then

$$\frac{\partial f}{\partial \overline{z}}(z_0)=0\qquad\text{and}\qquad f'(z_0)=\frac{\partial f}{\partial z}(z_0)=2\frac{\partial u}{\partial z}(z_0).$$

Also, if we write $F(x,y)=f(z)$, then $F$ is differentiable in the sense of real variables, and $\det J_F(x_0,y_0)=|f'(z_0)|^2$.

**Proof.** The Cauchy–Riemann equations are equivalent to $\partial f/\partial\overline{z}=0$. Then $f'(z_0)=\frac{1}{2}(\frac{\partial f}{\partial x}(z_0)+\frac{1}{i}\frac{\partial f}{\partial y}(z_0))=\frac{\partial f}{\partial z}(z_0)$ and $\partial f/\partial z=2\partial u/\partial z$. For differentiability of $F$: with $H=(h_1,h_2)$, $h=h_1+ih_2$, the Cauchy–Riemann equations give

$$J_F(x_0,y_0)(H)=\Big(\frac{\partial u}{\partial x}-i\frac{\partial u}{\partial y}\Big)(h_1+ih_2)=f'(z_0)h,$$

identifying complex numbers with pairs of reals. Finally, by the Cauchy–Riemann equations,

$$\det J_F(x_0,y_0)=\frac{\partial u}{\partial x}\frac{\partial v}{\partial y}-\frac{\partial v}{\partial x}\frac{\partial u}{\partial y}=\Big(\frac{\partial u}{\partial x}\Big)^2+\Big(\frac{\partial u}{\partial y}\Big)^2=\Big|2\frac{\partial u}{\partial z}\Big|^2=|f'(z_0)|^2. \tag{4}$$

■

The converse:

**Theorem 2.4**

Let $f=u+iv$ on an open $\Omega$. If $u,v$ are continuously differentiable and satisfy the Cauchy–Riemann equations on $\Omega$, then $f$ is holomorphic on $\Omega$ and $f'(z)=\partial f/\partial z$.

**Proof.** Write

$$u(x+h_1,y+h_2)-u(x,y)=\frac{\partial u}{\partial x}h_1+\frac{\partial u}{\partial y}h_2+|h|\psi_1(h),\qquad v(x+h_1,y+h_2)-v(x,y)=\frac{\partial v}{\partial x}h_1+\frac{\partial v}{\partial y}h_2+|h|\psi_2(h),$$

with $\psi_j(h)\to 0$ as $|h|\to 0$, $h=h_1+ih_2$ (differentiability follows from continuity of the partials). Using the Cauchy–Riemann equations,

$$f(z+h)-f(z)=\Big(\frac{\partial u}{\partial x}-i\frac{\partial u}{\partial y}\Big)(h_1+ih_2)+|h|\psi(h),$$

with $\psi=\psi_1+i\psi_2\to 0$. So $f$ is holomorphic and $f'(z)=2\partial u/\partial z=\partial f/\partial z$. ■

## 2.3 Power series

The complex exponential $e^z = \sum_{n\geq 0} z^n/n!$ converges absolutely for every $z$ ($\sum |z|^n/n! = e^{|z|}$), uniformly on every disc. It is entire with termwise derivative $(e^z)' = \sum nz^{n-1}/n! = e^z$ (shown below). The geometric series $\sum z^n$ converges absolutely only for $|z| < 1$, with sum $1/(1-z)$, since $\sum_{n=0}^{N} z^n = \frac{1-z^{N+1}}{1-z}$.

A **power series** is an expansion $\sum_{n=0}^{\infty} a_n z^n$ with $a_n \in \mathbb{C}$ (5). If it converges absolutely at $z_0$, it converges for all $|z| \leq |z_0|$.

**Theorem 2.5**

Given a power series $\sum_{n=0}^{\infty} a_n z^n$, there exists $0 \leq R \leq \infty$ such that:
(i) If $|z| < R$ the series converges absolutely.
(ii) If $|z| > R$ the series diverges. Moreover, if we use the convention that $1/0 = \infty$ and $1/\infty = 0$, then $R$ is given by Hadamard's formula

$$1/R = \limsup |a_n|^{1/n}.$$

$R$ is the **radius of convergence** and $|z| < R$ the **disc of convergence**; $R = \infty$ for $e^z$ and $R = 1$ for the geometric series.

**Proof.** Let $L = 1/R \neq 0, \infty$ (these cases are easy). If $|z| < R$, choose $\epsilon > 0$ with $(L+\epsilon)|z| = r < 1$; for large $n$, $|a_n|^{1/n} \leq L + \epsilon$, so $|a_n||z|^n \leq r^n$ and the series converges. If $|z| > R$, similarly a subsequence of terms is unbounded, so the series diverges. ■

**Remark**

On the boundary $|z| = R$ either convergence or divergence can occur.

Further entire power series are

$$\cos z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!}, \qquad \sin z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!},$$

agreeing with the usual functions for real $z$, and related to the exponential by the **Euler formulas**

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}, \qquad \sin z = \frac{e^{iz} - e^{-iz}}{2i}.$$

**Theorem 2.6**

The power series $f(z) = \sum_{n\geq 0} a_n z^n$ is holomorphic in its disc of convergence, and $f'(z) = \sum_{n\geq 0} n a_n z^{n-1}$, with the same radius of convergence.

**Proof.** Since $\lim n^{1/n} = 1$, $\limsup |a_n|^{1/n} = \limsup |na_n|^{1/n}$, so $\sum a_n z^n$ and $\sum n a_n z^n$, hence $\sum n a_n z^{n-1}$, have the same radius of convergence (Hadamard's formula).

Let $g(z) = \sum n a_n z^{n-1}$, let $R$ be the radius of convergence, and $|z_0| < r < R$. Write $f(z) = S_N(z) + E_N(z)$ with $S_N(z) = \sum_{n=0}^{N} a_n z^n$, $E_N(z) = \sum_{n=N+1}^{\infty} a_n z^n$. If $|z_0 + h| < r$,

$$\frac{f(z_0+h) - f(z_0)}{h} - g(z_0) = \left(\frac{S_N(z_0+h) - S_N(z_0)}{h} - S_N'(z_0)\right) + (S_N'(z_0) - g(z_0)) + \frac{E_N(z_0+h) - E_N(z_0)}{h}.$$

Since $a^n - b^n = (a-b)(a^{n-1} + a^{n-2}b + \cdots + b^{n-1})$,

$$\left|\frac{E_N(z_0+h) - E_N(z_0)}{h}\right| \leq \sum_{n=N+1}^{\infty} |a_n| \left|\frac{(z_0+h)^n - z_0^n}{h}\right| \leq \sum_{n=N+1}^{\infty} |a_n| n r^{n-1},$$

the tail of a convergent series ($g$ converges absolutely on $|z| < R$). Given $\epsilon > 0$, choose $N_1$ making this $< \epsilon$ for $N > N_1$, and $N_2$ with $|S_N'(z_0) - g(z_0)| < \epsilon$ for $N > N_2$. Fix $N > \max(N_1, N_2)$; as $S_N$ is a polynomial, $|\frac{S_N(z_0+h)-S_N(z_0)}{h} - S_N'(z_0)| < \epsilon$ for $|h| < \delta$. Then $|\frac{f(z_0+h)-f(z_0)}{h} - g(z_0)| < 3\epsilon$ for $|h| < \delta$. ■

**Corollary 2.7**

A power series is infinitely complex differentiable in its disc of convergence, the derivatives being termwise.

A power series centered at $z_0$ is $f(z) = \sum a_n(z - z_0)^n$; with $g(w) = \sum a_n w^n$, $f(z) = g(z - z_0)$, so everything about $g$ transfers, e.g. $f'(z) = \sum n a_n(z - z_0)^{n-1}$.

**Definition (analytic function)**

$f$ on an open $\Omega$ is **analytic** at $z_0$ if $f(z) = \sum a_n(z - z_0)^n$ near $z_0$ for a power series with positive radius; analytic on $\Omega$ if so at every point.

By Theorem 2.6, analytic functions are holomorphic; the converse (Chapter 2) justifies using the two terms interchangeably.

## 3 Integration along curves

A **parametrized curve** is a map $z : [a, b] \to \mathbb{C}$. It is **smooth** if $z'$ exists, is continuous and $z'(t) \neq 0$ on $[a, b]$, one-sided at the end-points:

$$z'(a) = \lim_{h \to 0,\, h>0} \frac{z(a+h) - z(a)}{h}, \qquad z'(b) = \lim_{h \to 0,\, h<0} \frac{z(b+h) - z(b)}{h}.$$

It is **piecewise-smooth** if $z$ is continuous and smooth on each $[a_k, a_{k+1}]$ for some $a = a_0 < \cdots < a_n = b$. Parametrizations $z : [a, b] \to \mathbb{C}$, $\tilde{z} : [c, d] \to \mathbb{C}$ are **equivalent** if $\tilde{z}(s) = z(t(s))$ for a $C^1$ bijection $t : [c, d] \to [a, b]$ with $t'(s) > 0$ (orientation preserved). The class determines a curve $\gamma \subset \mathbb{C}$, the image with its orientation; $\gamma^-$ is reversed, e.g. $z^-(t) = z(b + a - t)$. The **end-points** $z(a)$, $z(b)$ are parametrization-independent. $\gamma$ is **closed** if $z(a) = z(b)$, **simple** if $z(t) \neq z(s)$ for $s \neq t$ (for closed curves, except $\{s, t\} = \{a, b\}$). A *curve* means a piecewise-smooth curve.

$C_r(z_0)$ has the **positive** (counterclockwise) **orientation** $z(t) = z_0 + re^{it}$, $t \in [0, 2\pi]$, and the negative one $z_0 + re^{-it}$; $C$ denotes a positively oriented circle.

**Definition (integral along a curve; length)**

For a smooth curve $\gamma$ parametrized by $z : [a, b] \to \mathbb{C}$ and $f$ continuous on $\gamma$,

$$\int_\gamma f(z)\, dz = \int_a^b f(z(t))z'(t)\, dt.$$

For piecewise-smooth $\gamma$, $\int_\gamma f(z)\, dz = \sum_{k=0}^{n-1} \int_{a_k}^{a_{k+1}} f(z(t))z'(t)\, dt$. The **length** is $\text{length}(\gamma) = \int_a^b |z'(t)|\, dt$, summed over the smooth parts.

These are independent of the parametrization: if $\tilde{z}$ is equivalent to $z$, by change of variables and the chain rule

$$\int_a^b f(z(t))z'(t)\, dt = \int_c^d f(z(t(s)))z'(t(s))t'(s)\, ds = \int_c^d f(\tilde{z}(s))\tilde{z}'(s)\, ds.$$

**Proposition 3.1**

Integration of continuous functions over curves satisfies the following properties:
(i) It is linear: if $\alpha, \beta \in \mathbb{C}$, then $\int_\gamma (\alpha f(z) + \beta g(z))\,dz = \alpha \int_\gamma f(z)\,dz + \beta \int_\gamma g(z)\,dz$.
(ii) If $\gamma^-$ is $\gamma$ with the reverse orientation, then $\int_\gamma f(z)\,dz = -\int_{\gamma^-} f(z)\,dz$.
(iii) One has the inequality $\left|\int_\gamma f(z)\,dz\right| \le \sup_{z\in\gamma}|f(z)| \cdot \text{length}(\gamma)$.

**Proof.** (i) follows from linearity of the Riemann integral.

(ii) [Supplement]: with $z^-(t) = z(a+b-t)$, $\int_a^b f(z^-(t))(z^-)'(t)\,dt = -\int_a^b f(z(a+b-t))z'(a+b-t)\,dt = -\int_a^b f(z(s))z'(s)\,ds$. (iii): $\left|\int_\gamma f(z)\,dz\right| \le \sup_{t\in[a,b]}|f(z(t))| \int_a^b |z'(t)|\,dt \le \sup_{z\in\gamma}|f(z)| \cdot \text{length}(\gamma)$. ■

A **primitive** for $f$ on $\Omega$ is a function $F$ holomorphic on $\Omega$ with $F'(z) = f(z)$ for all $z \in \Omega$.

**Theorem 3.2**

If a continuous function $f$ has a primitive $F$ in $\Omega$, and $\gamma$ is a curve in $\Omega$ that begins at $w_1$ and ends at $w_2$, then $\int_\gamma f(z)\,dz = F(w_2) - F(w_1)$.

**Proof.** If $\gamma$ is smooth with parametrization $z : [a, b] \to \mathbb{C}$, $z(a) = w_1$, $z(b) = w_2$, then by the chain rule and the fundamental theorem of calculus

$$\int_\gamma f(z)\,dz = \int_a^b F'(z(t))z'(t)\,dt = \int_a^b \frac{d}{dt}F(z(t))\,dt = F(z(b)) - F(z(a)).$$

If $\gamma$ is piecewise-smooth we get a telescoping sum: $\sum_{k=0}^{n-1}[F(z(a_{k+1})) - F(z(a_k))] = F(z(b)) - F(z(a))$. ■

**Corollary 3.3**

If $\gamma$ is a closed curve in an open set $\Omega$, and $f$ is continuous and has a primitive in $\Omega$, then $\oint_\gamma f(z)\,dz = 0$.

Immediate, as the end-points coincide. E.g. $1/z$ has no primitive in $\mathbb{C} - \{0\}$: for the unit circle $z(t) = e^{it}$,

$$\oint_C f(z)\,dz = \int_0^{2\pi} \frac{ie^{it}}{e^{it}}\,dt = 2\pi i \neq 0.$$

This innocent calculation lies at the heart of the theory.

**Corollary 3.4**

If $f$ is holomorphic in a region $\Omega$ and $f' = 0$, then $f$ is constant.

**Proof.** Fix $w_0 \in \Omega$ and join it to $w$ by a curve $\gamma$ ($\Omega$ connected). By Theorem 3.2, $f(w) - f(w_0) = \int_\gamma f'(z)\,dz = 0$. ■